WEEK 2 · DAY 4 · OVERVIEW

Slopes and Butterflies



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ABOUT 45 MIN WITH EXERCISES / 8 SECTIONS / 4 EXERCISES

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The Desk Lead's Note

The morning note describes the curve in shorthand: 2s10s, 5s30s, 3m10y, and two flies. Give me all five for 2 October 2026 with the sign of each written down, so nobody reads one backwards, and put them in marketdata, tied to Excel. Check our 2s10s against FRED's own. Say where 2s10s and the 2s5s10s fly sit in their history. Give me the face amounts at which a 2s5s10s fly's three DV01s net to zero, from your bondmath. Last, keep 2026's slopes in a file under Git, so a rerun that moves a number shows up. Describe the shape; never say what it means for what comes next.

Your Route Through Day 4

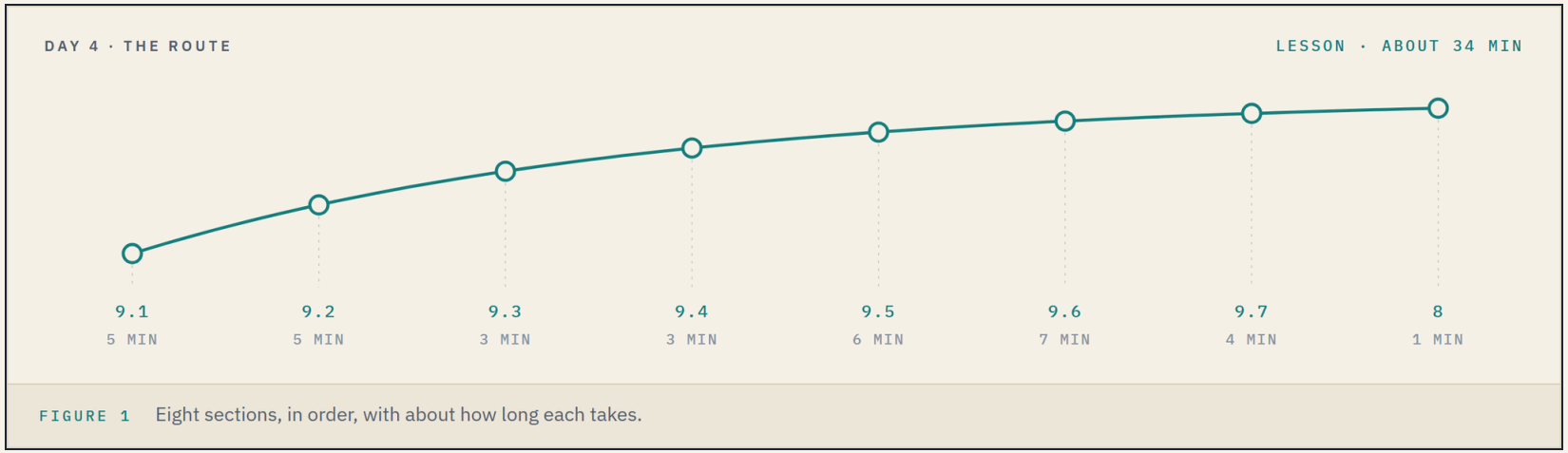


FIGURE 1 Eight sections, in order, with about how long each takes.

	SECTION	ABOUT
9.1	Slopes: 2s10s, 5s30s, 3m10y, With Signs	5 min
9.2	Butterflies: 2s5s10s and 5s10s30s, and the Other Sign	5 min
9.3	**`slope_bp`** and **`butterfly_bp`**: the Convention in the Docstring	3 min
9.4	Live: FRED's **`T10Y2Y`** Against Your 2s10s	3 min

	SECTION	ABOUT
9.5	Slopes and Flies Over Time	6 min
9.6	DV01-Neutral Fly Weights, From Your **`bondmath`**	7 min
9.7	2026's Slopes as a Data File Under Git	4 min
8	Bring It Home	1 min

Time. About 34 minutes for the lesson, then about 12 for the four exercises.

9.1 Slopes: 2s10s, 5s30s, 3m10y, With Signs

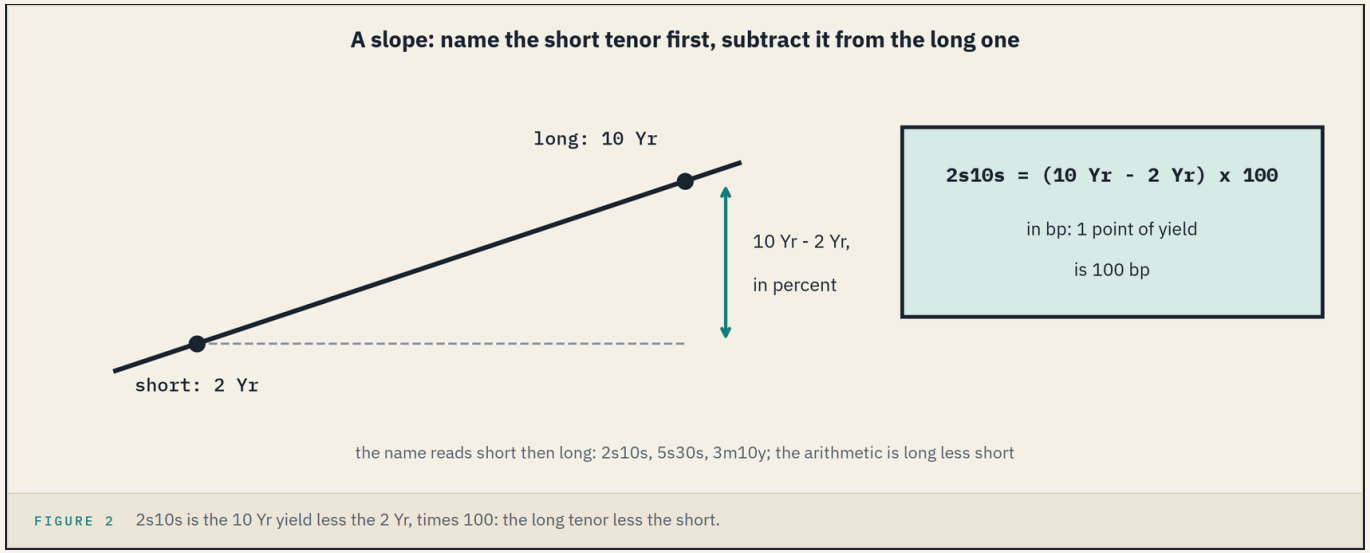


FIGURE 2 2s10s is the 10 Yr yield less the 2 Yr, times 100: the long tenor less the short.

EXCEL SIDE BY SIDE

JOB	EXCEL
2s10s on the last row	`=(M2941-I2941)*100`
2s10s on every row	`=(M2-I2)*100` filled to row 2941

2S10S BY HAND

```
two_tens_bp = (treasury.loc['2026-10-02', '10 Yr'] - treasury.loc['2026-10-02', '2 Yr']) * 100
```

```
2s10s on 2026-10-02: 45.00 bp
```

A slope is long less short, in bp: +45 bp on 2 October 2026, -108 bp at its lowest, and the sign is the direction of the line.

9.2 Butterflies: 2s5s10s and 5s10s30s, and the Other Sign

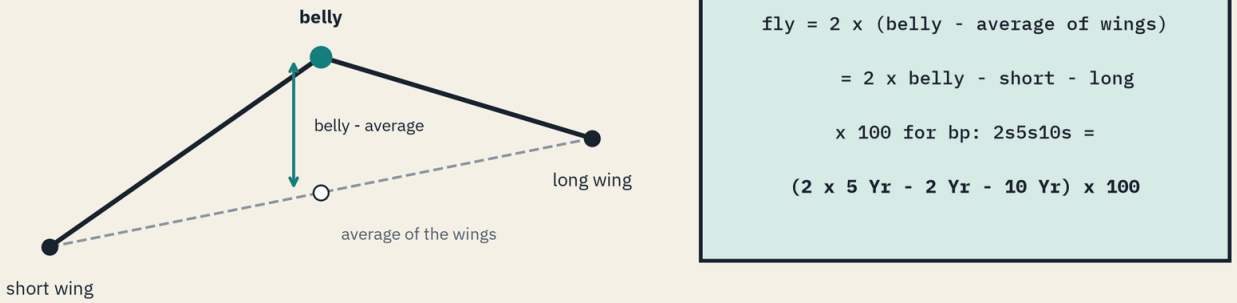


FIGURE 4 A fly is twice the belly less both wings, times 100: the belly sits above the wings' average.

EXCEL SIDE BY SIDE

JOB	EXCEL
2s5s10s	`=(2*K2-I2-M2)*100` filled down
5s10s30s	`=(2*M2-K2-O2)*100` filled down

THE OTHER SIGN

```
other_sign = -flies_by_hand['2s5s10s']
```

```
2s5s10s on 2026-10-02: +1.00 bp here, -1.00 bp on the other sign
```

Five numbers for 2 October 2026, each equal to Excel's on all 2,940 rows: the note's first ask.

9.3 slope_bp and butterfly_bp: the Convention in the Docstring

THE FUNCTION THAT CARRIES THE SIGN

```
def slope_bp(curve: pd.DataFrame, short: str, long: str) -> pd.Series:
    Units: yields in percent in, bp out. Convention: (curve[long] - curve[short]) x 100, named short then
def butterfly_bp(curve: pd.DataFrame, short: str, belly: str, long: str) -> pd.Series:
    Units: yields in percent in, bp out. Convention: (2 x belly - short - long) x 100, named short, belly,
```

SAME NUMBERS AS BY HAND

```
print('the same as by hand:', slopes.equals(slopes_by_hand), flies.equals(flies_by_hand))
print(md.slope_bp(treasury, '2 Yr', '10 Yr').name)
```

```
the same as by hand: True True
2 Yr to 10 Yr slope, bp
```

The sign now lives in the docstring and the tests pin it both ways: 38 passed, equal to Excel on all 190 reference rows.

9.4 Live: FRED's T10Y2Y Against Your 2s10s

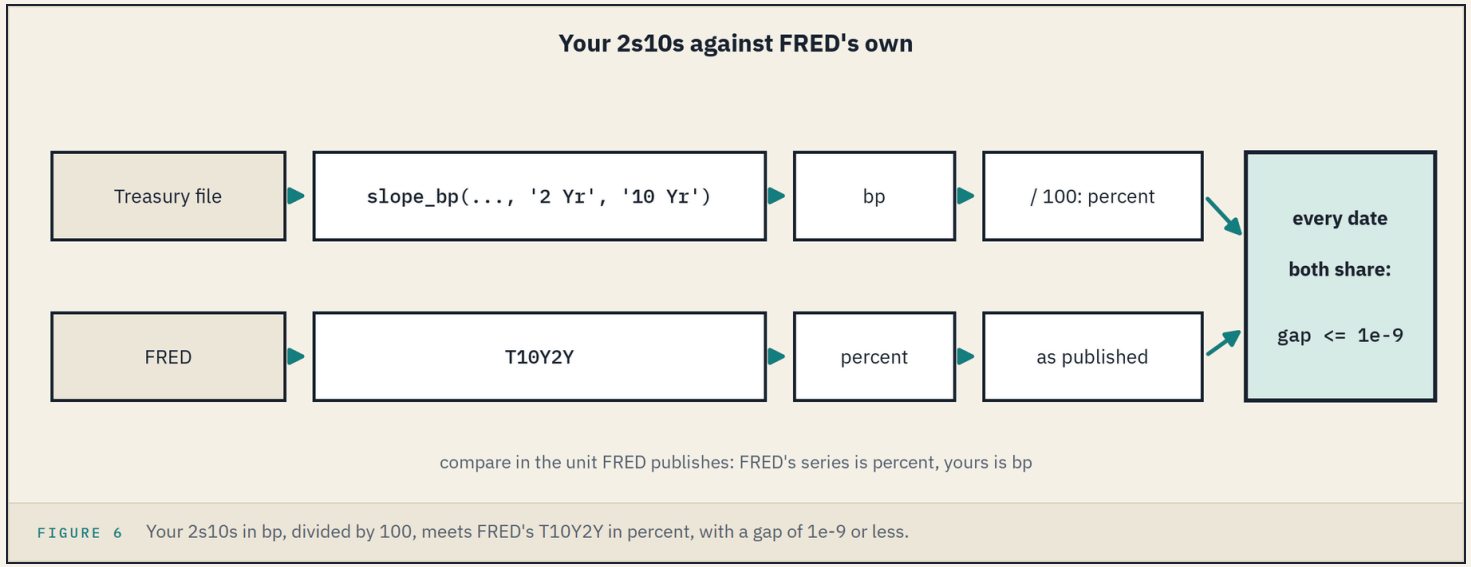


FIGURE 6 Your 2s10s in bp, divided by 100, meets FRED's T10Y2Y in percent, with a gap of 1e-9 or less.

WITH A KEY, IT PRINTS

With a key, the cell prints **T10Y2Y**'s rows since 2015 and its last date, later than the Treasury file's 2 October 2026 once FRED has published newer days, then the dates both share and the largest gap.

THE LIVE CELL

```
live_t10y2y = md.fred_series('T10Y2Y', fred_key, start='2015-01-01').dropna()
live_gap = (live_t10y2y[live_common] - slopes.loc[live_common, '2s10s'] / 100).abs().max()
```

With a key, your 2s10s divided by 100 equals FRED's `T10Y2Y` on every date the two share; without one, nothing else in the notebook changes.

9.5 Slopes and Flies Over Time

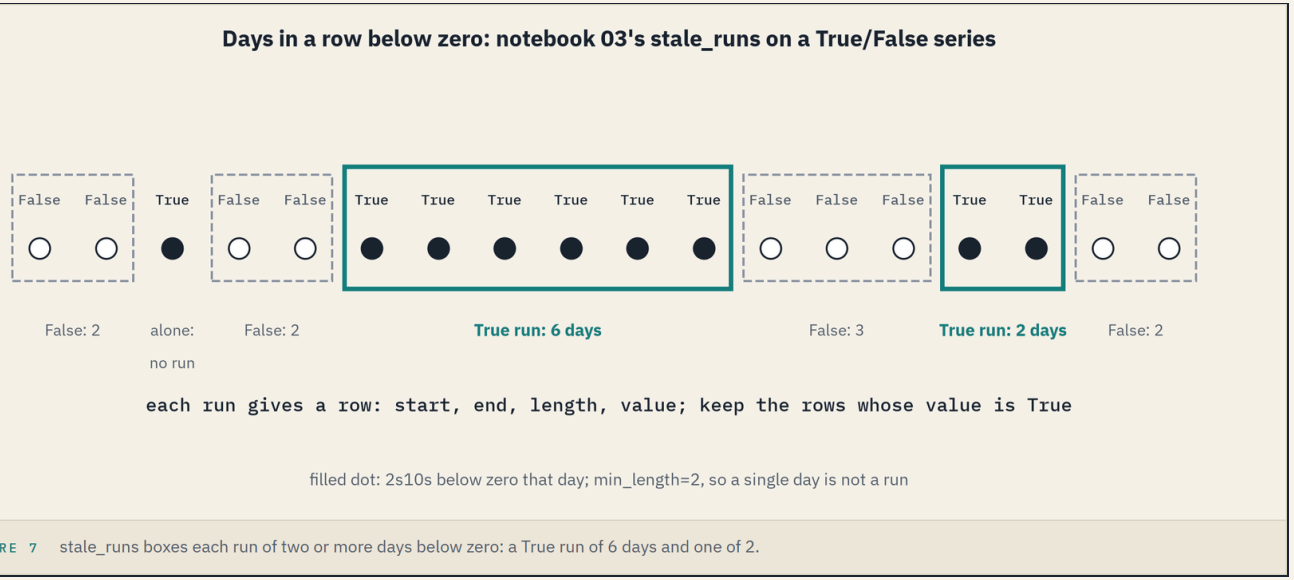


FIGURE 7 stale_runs boxes each run of two or more days below zero: a True run of 6 days and one of 2.

EXCEL SIDE BY SIDE

JOB	EXCEL
Days below zero	`=COUNTIF(E2:E2941,"<0")`
Days exactly zero	`=COUNTIF(E2:E2941,0)`
The lowest	`=MIN(E2:E2941)`

THE LONGEST RUN

```
longest = negative_runs.loc[negative_runs['length'].idxmax()]
```

```
3 runs below zero; the longest 537 market days, 2022-07-06 to 2024-08-26
```

One number on one date; its history says where it sits.

9.6 DV01-Neutral Fly Weights, From Your bondmath

THREE PAR BONDS

```
for tenor, years in [('2 Yr', 2), ('5 Yr', 5), ('10 Yr', 10)]:
    dv01_by_tenor[tenor] = bondmath.dv01(settlement, maturity, yield_pct, yield_pct)
```

```
2 Yr: price 100.000000, DV01 0.018848 per 100 of face per bp
5 Yr: price 100.000000, DV01 0.043693 per 100 of face per bp
10 Yr: price 100.000000, DV01 0.076925 per 100 of face per bp
```

EXCEL SIDE BY SIDE

JOB	EXCEL
5-year price	`=PRICE(DATE(2026,10,2),DATE(2031,10,2),5.06/100,5.06/100,100,2,0)`
5-year accrued	`=100*5.06/100/2*COUPDAYBS(...)/COUPDAYS(...)`
5-year modified duration	`=MDURATION(DATE(2026,10,2),DATE(2031,10,2),5.06/100,5.06/100,2,0)`
5-year DV01	`=C3*(C1+C2)/10000` in C4
2-year and 10-year DV01	the same in B4 and D4

The Wings Net the Belly

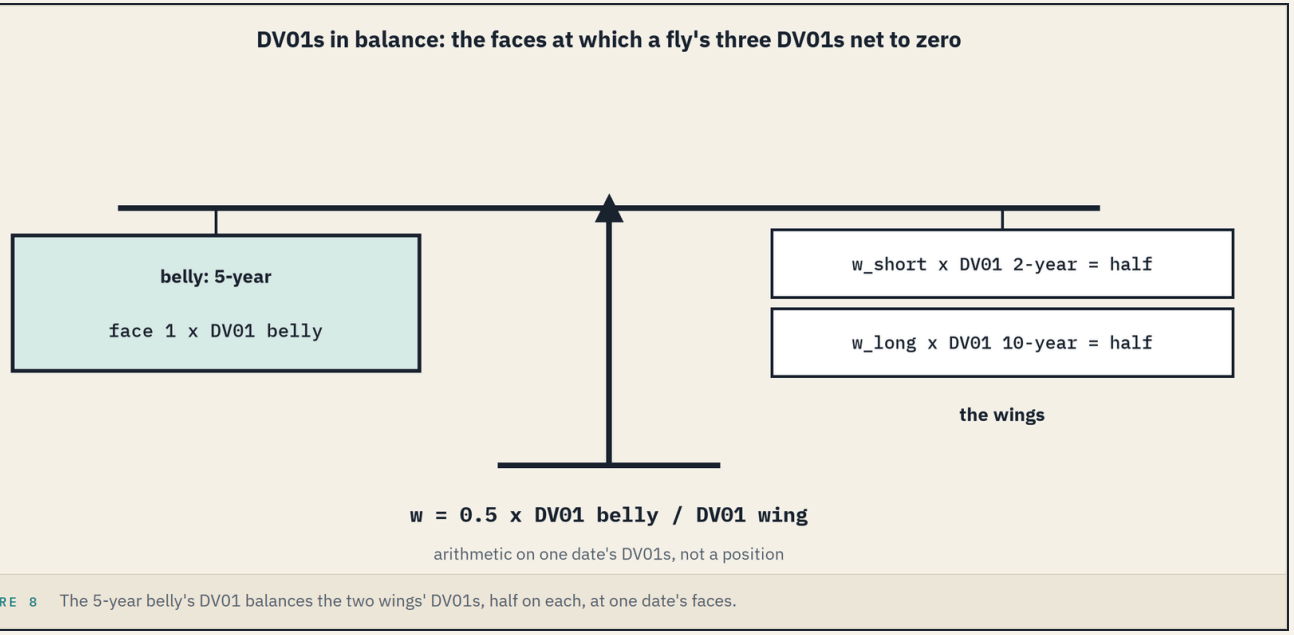


FIGURE 8 The 5-year belly's DV01 balances the two wings' DV01s, half on each, at one date's faces.

EXCEL SIDE BY SIDE

JOB	EXCEL
2-year face	`=0.5*C4/B4`
10-year face	`=0.5*C4/D4`
The net	`=C4-B6*B4-C6*D4`

THE FACES, AND THE NET

```
print(f'faces per 1 of 5-year face: 2-year {w_short:.6f}, 10-year {w_long:.6f}')
print(f"belly {dv01_by_tenor['5 Yr']:.6f} = {short_dv01:.6f} + {long_dv01:.6f}; net {net:.1e}")
```

```
faces per 1 of 5-year face: 2-year 1.159049, 10-year 0.283995
belly 0.043693 = 0.021846 + 0.021846; net 0.0e+00
```

9.7 2026's Slopes as a Data File Under Git

WRITE THE FILE

```
slopes.loc['2026'].to_csv('slopes_2026.csv', float_format='%.2f', lineterminator='\n')
```

```
190 rows under the header
date,2s10s,5s30s,3m10y
2026-01-02,72.00,112.00,54.00
```

A SLIP, THEN THE DIFF

```
edited.loc['2026-09-30', '2s10s'] = -41.0
!git -C "{work}" diff slopes_2026.csv
```

```
--- a/slopes_2026.csv
+++ b/slopes_2026.csv
-2026-09-30,41.00,55.00,109.00
+2026-09-30,-41.00,55.00,109.00
```

One - line, the committed 41.00, and one + line, the slip: a hunk read as in Course 2's Cash Flows and Price From Yield.

Bring It Home

IN A POWERSHELL TERMINAL

```
# the work folder: use the name your setup cell printed
$work = "$env:TEMP\pfi_course3_09_k4w7q2xr"
Set-Location "$HOME\projects\my_bondmath"
Copy-Item "$work\marketdata.py", "$work\test_marketdata.py", "$work\slopes_2026.csv" .
python -m pytest -q test_marketdata.py
git status --short
git add marketdata.py test_marketdata.py slopes_2026.csv
git commit -m "Notebook 09: slope_bp, butterfly_bp, fly_weights, and slopes_2026.csv"
```

pytest shows 40 passed; `git status --short` shows the two files changed and `slopes_2026.csv` new.

You Can Now

- Compute a slope as long less short in bp, and read its sign as the direction of the curve.
- Compute a butterfly as twice the belly less both wings, and convert from the other desk's sign.
- Write a sign convention into a docstring and pin it with tests tied to Excel.
- Check your 2s10s against FRED's **`T10Y2Y`** in the unit FRED publishes.
- Count days and runs below zero, and say where a slope or fly sits in its own last year.
- Compute DV01-neutral fly weights from **`bondmath`**, as arithmetic, not a position.
- Keep a data file under Git, read a changed number in the diff, and restore it.

NEXT
WEEK 2 · DAY 5

Case Study: Three Years of Curve and Spreads

Every measure of the week in one table as of one date, tied to Excel and blind to later rows; then the same function describes three years of the curve and two synthetic spreads.